Prabhavit Gupta

Senior Software Engineer | Infrastructure Agent Systems, Autonomous Observability & AI Platform

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Executive Summary

- **Autonomous Infrastructure Agents & Orchestration:** Systems-oriented software engineer with 3 years commercial software experience architecting production AI agent runtimes, orchestration engines, and automated incident diagnosis and remediation loops for distributed systems.
- **Agent Platform, Knowledge Grounding & Tool Use:** Deep hands-on expertise building sandboxed tool execution harnesses, Model Context Protocol (MCP) servers, structured retrieval/knowledge systems, and quantitative eval frameworks (Langfuse, DeepEval) to ground agent reasoning deterministically in live system telemetry.
- **Distributed Backend Systems & Resilient Automation:** Production background in Python (AsyncIO, FastAPI), TypeScript, Docker, Kubernetes environments, and event-driven architectures, designing high-throughput services with rigorous observability and fault isolation.

Technical Skills & Core Competencies

- **AI Agent Systems & Orchestration:** Autonomous Agent Loops, Multi-Agent Coordination, Tool Calling, Model Context Protocol (MCP), Grounded Reasoning, Self-Correction Loops, Automated Incident Investigation & Remediation.
- **Agent Platform & Knowledge Systems:** Knowledge Graphs, Vector Retrieval / RAG, Context Engineering, Tool Verification Harnesses, Quantitative Evaluations (Langfuse, DeepEval), Prompt Regression Testing.
- **Distributed Systems & Backend Engineering:** Python (AsyncIO, FastAPI, PyTest), TypeScript, Node.js, Go fundamentals, RESTful APIs, GraphQL, Event-Driven Messaging (Kafka, NATS concepts), PostgreSQL, Redis, SQLite.
- **Infrastructure & Observability:** Docker, Kubernetes, Linux Internals & CLI, Prometheus & Grafana integration patterns, Distributed Tracing, GitOps / GitHub Actions CI/CD.

Professional Experience

Senior AI Agent Systems Engineer & Solutions Architect • *Independent Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid)

- Architected and deployed production autonomous AI agent systems that monitor, diagnose, and remediate operational failures across distributed customer infrastructure.
- Built a modular orchestration engine and tool-calling harness utilizing Model Context Protocol (MCP), enabling agents to query live system logs, execute diagnostic checks, and trigger verified remediation scripts safely.
- Implemented automated incident correlation pipelines synthesizing telemetry, error traces, and operational playbooks into dynamic context windows for LLM reasoning.
- Designed comprehensive quantitative evaluation harnesses using Langfuse, benchmarking agent decision accuracy, tool parameter validation, and hallucination rates across production failure scenarios.
- Authored clean, production-grade integration libraries in Python and TypeScript connecting agents to ticketing systems, monitoring alerts, and internal APIs.

Lead Developer & Platform Engineer • *FlowSpace.co*
Mar 2024 – Aug 2026 • London, UK (Remote)

- Led platform infrastructure and API development, scaling the core service to handle 4,500+ active enterprise users with high availability across European tech hubs.
- Re-architected core backend services from legacy infrastructure to Next.js and high-concurrency API backends, establishing zero-downtime deployment pipelines and strict data validation layers.
- Built automated asynchronous data ingestion pipelines collecting and transforming operational inventory from external APIs with automated fault isolation and retries.
- Engineered automated incident triage workflows and webhook integrations (OfficeRnD, Nexudus) ensuring real-time state synchronization across distributed booking resources.
- Implemented automated monitoring, logging, and error tracing, reducing operational mean-time-to-detection (MTTD) for service degradations.

Full-Stack Systems Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Developed backend microservices and automated workflows in Python and TypeScript, connecting heterogeneous client platforms to internal data storage layers.
- Built containerized test suites and CI/CD pipelines in Docker and GitHub Actions, enforcing automated schema

verification and integration testing prior to release.
- Partnered directly with engineering and operations teams to translate operational pain points into maintainable software architectures.

Key Engineering Projects

- **Autonomous Infrastructure Diagnostic Harness (Python + MCP + Langfuse):** Built an agentic diagnostic framework that automatically ingests system alerts, executes targeted diagnostic tool calls over MCP, verifies root-cause hypotheses against live telemetry, and outputs structured remediation plans.
- **Universal Multi-Source Telemetry Ingestion Pipeline (Python + AsyncIO + Docker):** Designed an asynchronous event ingestion engine capable of parallel parsing across 900+ structured and semi-structured external sources with schema enforcement and error quarantine.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy, International Studies & Economics** – Ca' Foscari University of Venice (2020 – 2024)
  Rigorous foundation in formal logic, epistemology, decision theory, and complex systems analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Legal Status & Availability:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy). UK / European remote aligned; available immediately.